

Degradation behavior of polycarbonate during melt processing with an aqueous solution of sodium hydroxide

Youngsoo Choi, Hyungsu Kim^{*}

Department of Chemical Engineering, Dankook University, 152 Jukjeon-ro, Yongin-si, Gyeonggi, 16890, South Korea

ARTICLE INFO

Keywords:

Polycarbonate
Hydrolysis
Controlled degradation
Rheological properties

ABSTRACT

In this study, we aimed to enhance the reactivity of polycarbonate (PC) with the epoxy group of styrene-co-glycidyl methacrylate copolymer (SGMA) by introducing aqueous solutions of sodium hydroxide during the melt processing of PC. This approach led to the generation of hydroxyl groups at the ends of PC molecules, facilitating the formation of PC-SGMA copolymer. The effectiveness of this method in improving the compatibility of PC and styrene-acrylonitrile (SAN) blends was successfully confirmed through blending experiments with SAN and SGMA. Additionally, we investigated the impact of alkali-promoted hydrolysis on the rheological properties of PC, apart from the functionalization of PC end groups. Our findings revealed that an increase in the concentration of sodium hydroxide resulted in a decrease in the complex viscosity of PC. However, beyond a certain concentration level, the viscosity exhibited an upward trend. Furthermore, when constructing van Gurp-Palmen plots for the modified PC, we observed a significant deviation in curve shape compared to pure PC or water-modified PC. This observation suggests that changes in the microstructure occur during the degradation process due to the interaction between PC molecules and sodium ions.

1. Introduction

Controlled degradation of polymers during melt compounding offers a versatile approach to manipulate the molecular weight and flow characteristics of the initial resin. One prominent example is the peroxide-induced degradation of polypropylene (PP) [1,2]. The end group composition of polycarbonate (PC) varies depending on the molecular weight modifier utilized during the polymerization process. In solution polymerization, *p*-tert-butylphenol is commonly employed as a molecular weight regulator, resulting in the absence of hydroxyl groups at the end of PC. Meanwhile, during melt polymerization via transesterification of bisphenol-A and diphenylcarbonate, the polymerized PC can possess phenoxy or hydroxy groups at the end, depending on the monomer ratio [3].

When the concentration of hydroxy groups at the end of PC molecules is low, achieving effective reactive compatibilization with other polymers may be challenging due to insufficient reactivity. PC molecules are susceptible to hydrolysis at temperatures ranging from 40 to 100 °C [4–7]. This mechanism involves nucleophilic attack on carbonate linkages by water, resulting in the formation of hydroxyl end groups in the cleaved molecules. The presence of alkali catalysts and various

temperature and pressure conditions can facilitate the conversion of PC into bisphenol-A and dimethyl carbonate, offering potential routes for chemical recycling [8–11].

In this study, our objective was to enhance the reactivity between PC and styrene-acrylonitrile-glycidyl methacrylate copolymer (SAG) by introducing a hydroxy group at the end of PC through the melt-mixing process using an aqueous alkali solution. We anticipated that the reaction between the epoxy group in the SAG copolymer and the hydroxy group in PC would facilitate the combination of PC and SAG, thus contributing to copolymer formation. By activating the PC end group with a hydroxy group, we aimed to improve the compatibility of commercially important PC/ABS blends, leveraging the favorable interaction between the styrene-acrylonitrile (SAN) component of ABS and SAG [12,13]. Achieving compatibility in PC/ABS blends is of significant commercial importance, as the phase structure and performance of these blends are highly sensitive to the processing conditions employed [14]. Therefore, successful compatibilization is vital to maintaining the desired characteristics and functionality of the blend throughout the production process.

In addition to investigating the compatibility enhancement, we also examined how the introduction of sodium ions during the melt

^{*} Corresponding author.

E-mail address: hkim@dku.edu (H. Kim).

processing affect the rheological properties of PC. We analyzed the changes in melt viscosity and viscoelasticity of PC under various concentrations of aqueous alkali solutions.

2. Experimental

2.1. Materials and sample preparation

The polycarbonate (PC) used in this study was Trirex 3020, obtained from Samyang Corporation (Korea). Its number average molecular weight (M_n) was determined to be 14,300 g/mol with a polydispersity of 2.5, as measured by gel permeation chromatography (GPC). The styrene-acrylonitrile-glycidyl methacrylate copolymer (SAG) with the designation SAG-002 was purchased from Fine-blend Polymer (China). The styrene-acrylonitrile copolymer (SAN) was provided by Lotte Advanced Materials (Korea) and has an acrylonitrile (AN) content of 24 wt%. The SAN has a M_n of 59,000 g/mol with a polydispersity of 2.2. Sodium hydroxide (NaOH) and tetrahydrofuran (THF) were purchased from Samchun Chemicals (Korea) and were used as received without further purification.

The melt mixing process to obtain modified PC (referred to as mPC) was conducted using an Internal Mixer (MKE RheoComp® System, Korea). The mixing operation was carried out at 240 °C with a rotor speed of 60 RPM for a duration of 10 min. The total amount of polymer resin used was 45 g. Various molar ratios of sodium ions to PC were achieved by adjusting the concentration of NaOH/H₂O solutions. Prior to the melt processing step, the resins were dried in a vacuum oven at 70 °C for 24 h. For the evaluation of rheological properties, samples obtained from the mixer were compressed using a hot press (Carver Laboratory Press, USA) at 260 °C and 9.5 tons of pressure. This resulted in disk-shaped specimens with a thickness of 2 mm and a diameter of 25 mm. In preparation for atomic force microscopy (AFM) analysis, a film with a thickness of 1 mm was created using the hot press at 260 °C and a pressure of 13.5 tons. After molding, all specimens were cooled to room temperature in ambient air. Specimens for tensile testing, following ASTM 648 specifications, were fabricated using an injection molding machine (Haake Minijet Pro Piston Injection Molding System, USA). The cylinder temperature was set to 260 °C, while the mold temperature was set to 160 °C. A pressure of 700 bar was applied for 5 s, followed by a subsequent pressure of 450 bar maintained for 5 s.

2.2. Measurements

The storage modulus (G'), loss modulus (G''), and complex viscosity (η^*) of the samples were measured using a plate-plate type rotational rheometer (Anton Paar, MCR500). The measurements were conducted at 240 °C within a frequency range of 0.1–100 rad/s. The gap size between the plates was set at 2 mm, and a strain of 1% was applied to ensure linear viscoelasticity.

To determine the average molecular weight, the polycarbonate resin was dissolved in tetrahydrofuran (THF) solvent (4 mL) and analyzed using the Gel Permeation Chromatography (GPC) method. Specifically, 20 mg of the PC resin was used, and the analysis was performed with a Tosho EcoSEC HLC-8420 (Japan) GPC instrument.

The presence of OH groups in the PC was verified using a UV–visible spectrophotometer (UV VIS, OPTIZEN 2120UV, Korea). Preliminary tests were conducted on standard solutions containing varying amounts of bisphenol A. Through these tests, it was confirmed that the absorbance at 289 nm, corresponding to the OH group, changes in relation to the concentration of bisphenol A.

The morphology of the PC/SGMA blend was analyzed using a field emission scanning electron microscope (FE-SEM, Hitachi S-4700, Japan). To prepare the specimens for imaging, the surface was coated with platinum after fracturing the sample in liquid nitrogen. The SEM images were then processed using ImageJ software (National Institute of Health, USA) to measure the volume average radius (R_v) of the dispersed

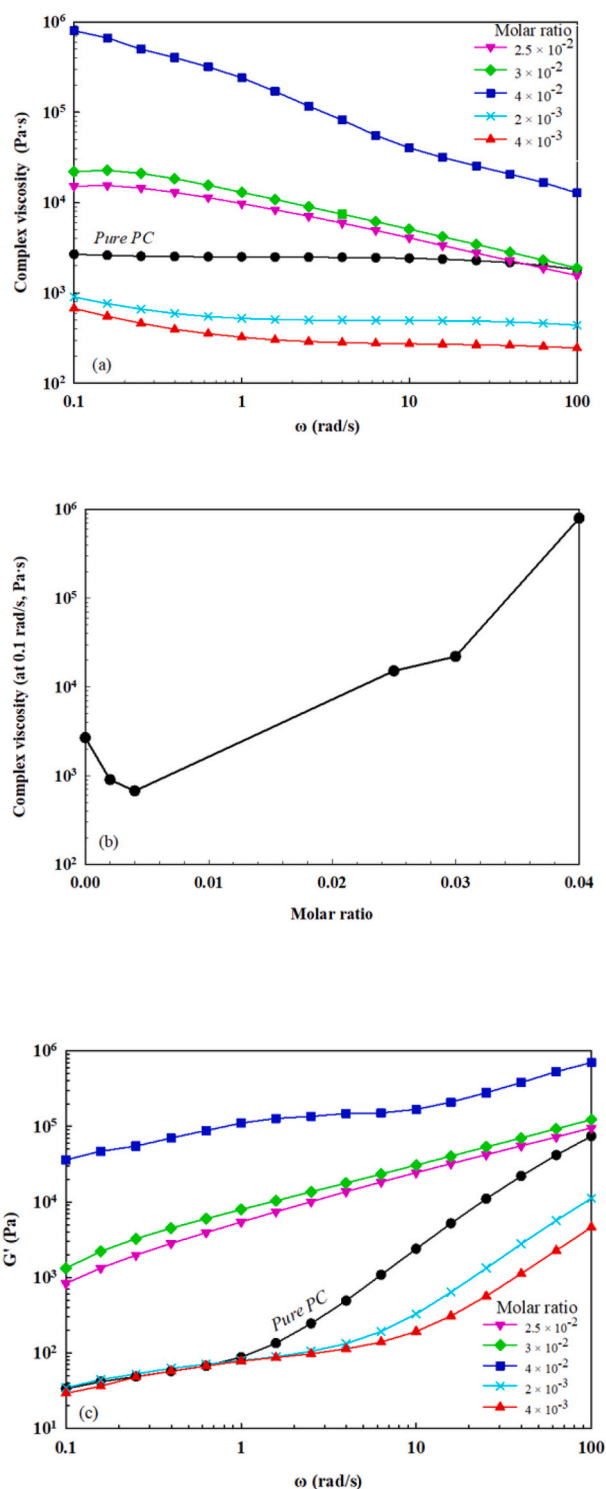


Fig. 1. Effect of molar ratio of sodium and PC on the rheological properties of PC at 240 °C: (a) complex viscosity vs. angular frequency, (b) complex viscosity (at 0.1 rad/s) vs. molar ratio, (c) storage modulus vs. angular frequency.

phase.

For the analysis of the phase structure, an Atomic Force Microscope (AFM, XE 150, Park Systems, Korea) was utilized. The measurement was performed in tapping mode with a scan rate of 0.5 Hz within a range of 5 μ m in the X-axis and Y-axis. To obtain a smooth surface, the molded film was heat-treated at 200 °C for 2 h and subsequently air-cooled. It is worth noting that a smooth surface improves the resolution of phase

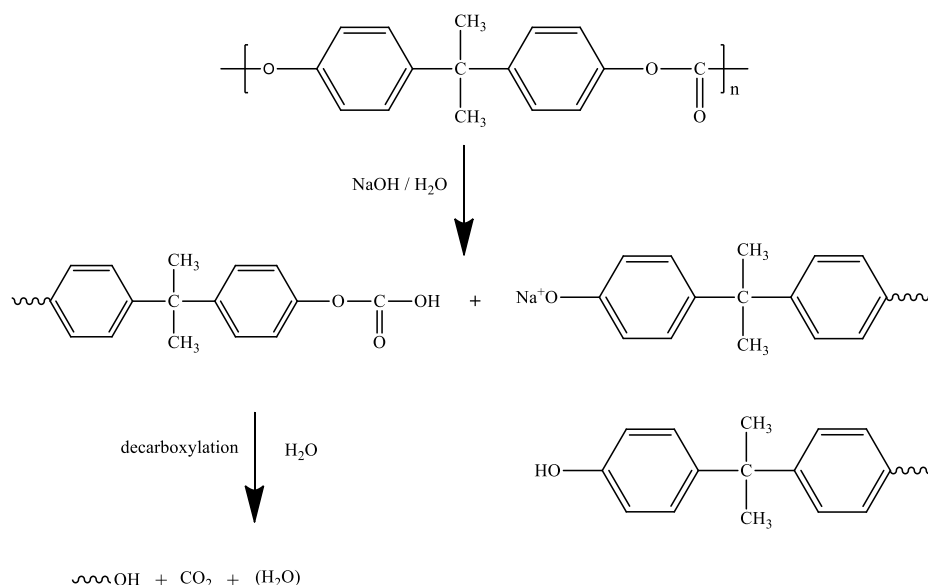


Fig. 2. Mechanism of PC degradation by NaOH with H₂O. Here, other compounds such as phenol and bisphenol by-products were excluded [9,10].

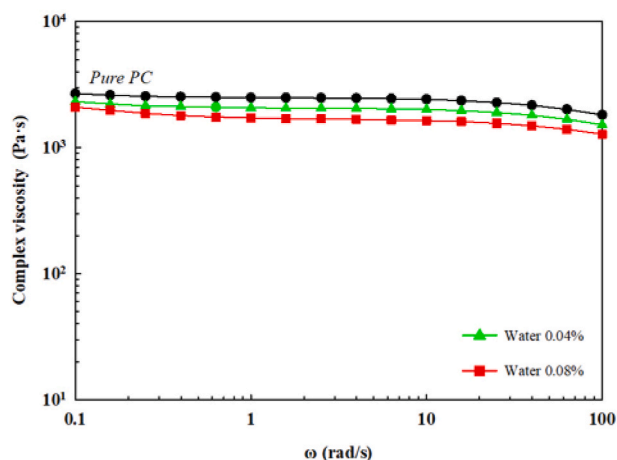


Fig. 3. Effect of water content on the melt viscosity of PC.

structures in AFM studies [15]. The as-molded film was investigated to ensure no significant change in the size of ionic domains, considering the possibility of ion aggregate reconstruction during thermal treatment.

The dispersion state and location of ion clusters present in the PC domain in the mPC/SAN/SGMA blend were investigated using transmission electron microscopy (FE-TEM, JEOL JEM-2100F). Additionally, the characteristics of these ion clusters were analyzed through Energy Dispersive X-ray Spectroscopy (EDX), utilizing the Oxford AZtec X-Max system. Prior to TEM observation, ultrathin sections approximately 75–90 nm thick were prepared using an ultramicrotome system (PT PC Ultramicrotome & Photographic) at room temperature.

The elongation at failure of the specimens was measured using Lloyd's Universal Testing Machine (LR5K plus, England). The tensile strain rate was set to 1.667 mm/s, and the reported values represent the mean of measurements obtained from 5 specimens.

3. Results and discussion

Fig. 1(a) illustrates the impact of the molar ratio of sodium to PC on the complex viscosity of PC. It was observed that as the concentration of sodium hydroxide increased, the complex viscosity of PC decreased.

Table 1

Molecular weight information for pure PC and modified PC.

	Pure PC	2×10^{-3a}	4×10^{-3a}	Water 0.04%	Water 0.08%
M_n	14,300	12,676	12,162	12,134	11,306
M_w	35,873	27,206	25,525	35,976	35,518
M_w/M_n	2.5	2.1	2.1	2.9	3.1

^a The number indicates the molar ratio of NaOH to PC.

However, somewhat unexpectedly, it started to increase when the molar ratio exceeded 0.004. To provide a clearer comparison, Fig. 1(b) depicts the viscosity measured at 0.1 rad/s as a function of the molar ratio.

Fig. 2 outlines the mechanism by which the nucleophilic attack of hydroxide ions, predominantly derived from the ionization of NaOH in the aqueous phase, leads to chain scission at the carbonyl carbon, resulting in a reduction in the molecular weight of PC and subsequently the melt viscosity. To compare the impact of hydrolysis without NaOH on the PC melt viscosity, identical amounts of water required for the 2 M ratios of 2×10^{-3} and 4×10^{-3} were employed during the melt processing of PC. Fig. 3 demonstrates that the melt viscosity of PC still decreased due to hydrolysis, but to a lesser extent than when NaOH was present. This result emphasizes the significance of Na ions in the modification process. It is important to note that the degradation products include hydroxy-terminated PC and sodium salts of the remaining fragments. To determine the amount of free NaOH remaining unconsumed in the reaction, precipitation of mPC was conducted by adding water to the mPC/THF solution. After filtration of mPC, it was confirmed that the residual solution did not react with phenolphthalein indicator, indicating the absence of NaOH loss.

Additionally, mPC with a higher viscosity than pristine PC exhibited insolubility in THF and could not be analyzed using GPC. Based on these observations, we propose that the rheological properties of mPC, under specific ion concentration conditions, are influenced by the formation of ionic crosslinking. This hypothesis is supported by Fig. 1(c), where the storage moduli of mPC demonstrate a significant increase, indicating an enhanced elastic behavior of the modified resin.

Table 1 provides an overview of the variations in the molecular weight of PC resulting from modification with aqueous solutions of NaOH or water alone. The results clearly demonstrate significant reductions in both the number average molecular weight (M_n) and weight average molecular weight (M_w) when NaOH is added. Additionally, the

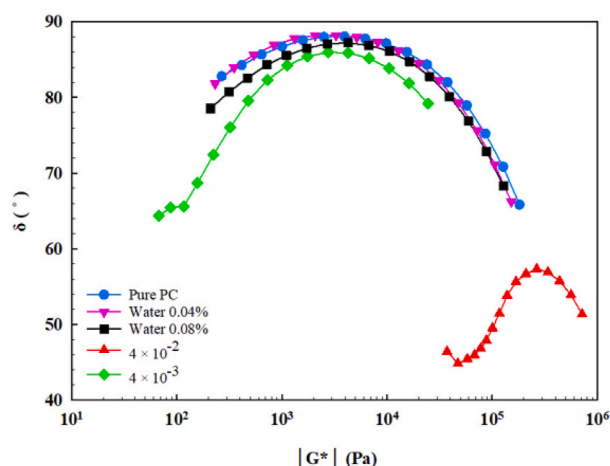


Fig. 4. Van Gorp-Palmen plots of phase angle (δ) vs. complex modulus ($|G^*|$) at 240 °C for pure PC and modified PC.

molecular weight distribution (MWD) decreases compared to pure PC. Conversely, when PC is modified solely with water, there is a decrease in M_n but no significant impact on M_w . Notably, the MWD value is higher than that of pure PC in this case. These findings indicate that while the decomposition of PC molecules can occur with water alone, the presence of sodium ions in the form of NaOH intensifies the degree of decomposition and alters the molecular weight distribution of PC molecules. It is worth noting that the sodium ions not only accelerate the decomposition of PC molecules but also cause structural transformations of PC depending on their concentration.

The van Gorp-Palmen plot, which relates the phase angle (δ) to the complex shear modulus ($G^*(\omega)$), is a valuable tool for analyzing the molecular structure and relaxation behavior of polymers [16,17]. This plot has been utilized to gain insights into the formation of long-chain branching resulting from reactive extrusion or electron beam irradiation of polypropylenes [18,19]. In our study, we constructed the van Gorp-Palmen plot for the modified samples (Fig. 4). The results indicate that the curves for water-modified PC and pure PC overlap, suggesting no significant variation in the molecular structure. However, for the samples modified with aqueous solutions of NaOH, a noticeable variation of the curve shape is observed. Specifically, the high-viscosity sample with a higher molar ratio shows significant deviation from the pristine PC and a substantial reduction in the loss factor. This comparison yields significant implications as the distinct shape of the van

Gorp-Palmen plot suggests a change in the molecular architecture of the PC. When modifying PC with water alone, chain scission of the molecules predominates without any alterations in the molecular structure. In contrast, NaOH-induced modification implies the potential occurrence of crosslinking in PC, with varying degrees of crosslinking and the formation of ion aggregates that interact with the PC fragments. However, caution is required when interpreting these results, and further investigations are necessary to determine the exact structure of mPC. Nonetheless, based on the topography of the van Gorp-Palmen plot, we presume that structural changes in PC have taken place during the sodium-incorporated modification process.

To verify the existence of ion aggregates, we analyzed the surface of polymer film by using AFM techniques that are effective in resolving the ionic domains in the polymer matrix [20]. Comparing the phase images shown in Fig. 5, the surface of PC is relatively featureless, while black dots (see circles in the figure) are seen in mPC due to the existence of Na ion aggregates. The size of the domain ranges from 140 to 150 nm in diameter, which is larger than the ionic clusters of approximately 5–10 nm found in typical ionomers such as Surlyn and Nafion [20]. Although mPC belongs to ion containing polymers, a direct comparison with ionomers may not be appropriate given the differences in manufacturing methods. To elucidate the structure of mPC more clearly, further analysis, including small angle X-ray scattering (SAXS) analysis, is underway.

To ensure the generation of hydroxy end-groups through hydrolysis,

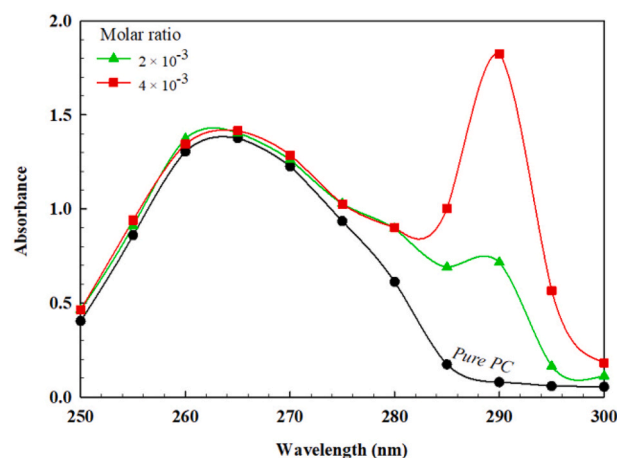


Fig. 6. UV absorption spectra for pure PC and modified PC.

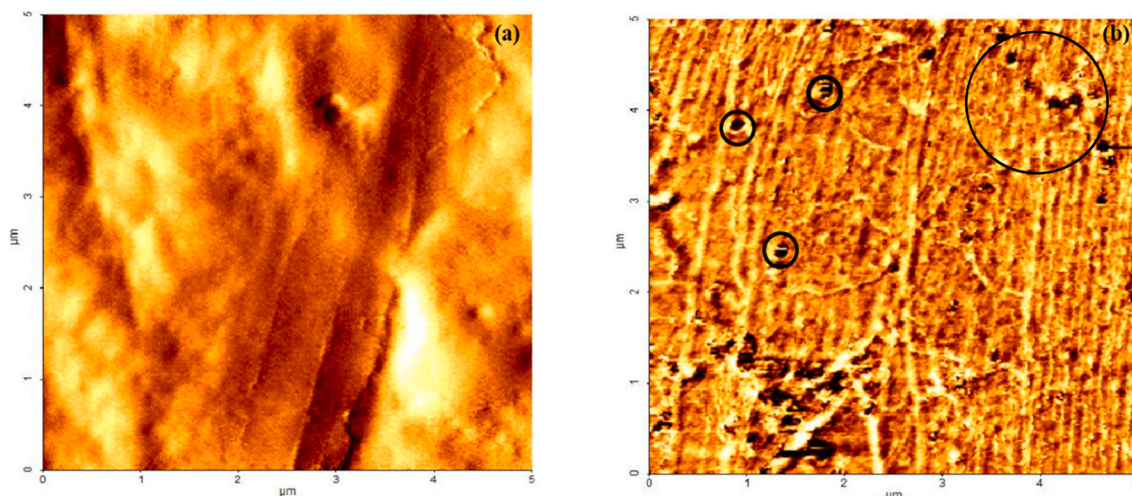


Fig. 5. Phase images of PC (a) and mPC (molar ratio = 4×10^{-3}) (b) in AFM. No height data are shown. Ionic clumps are indicated with circles in (b).

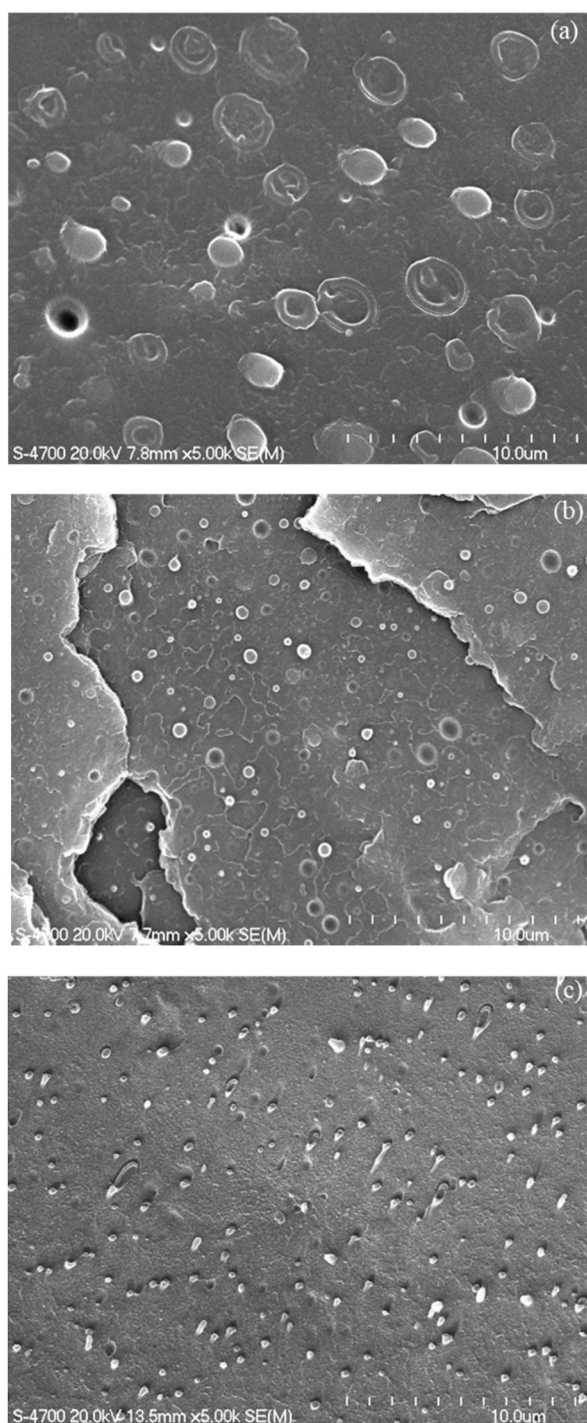


Fig. 7. SEM images for (a) PC/SAN/SAG ($R_v = 0.98 \mu\text{m}$), (b) PC(water)¹/SAN/SAG ($R_v = 0.30 \mu\text{m}$), and (c) mPC²/SAN/SAG ($R_v = 0.19 \mu\text{m}$). The blend ratio of PC/SAN is 20/80. For all blends, 5 wt% of SAG was added. ¹PC was modified with 0.08 wt% of water without NaOH. ²PC was modified at the molar ratio of 4×10^{-3} with 0.08 wt% of water.

we conducted analysis on the samples using a UV spectrometer. The corresponding results are presented in Fig. 6. The data obtained clearly indicate the absence of absorbance around 289 nm in the PC utilized in this study, suggesting a minimal presence of hydroxy groups [21]. As the concentration of NaOH increases, the intensity of the associated peak rises, providing further confirmation that phenolic end-groups were indeed produced during the melt processing of PC with aqueous solutions of NaOH.

To improve the compatibility of the PC/SAN blend, we explored the potential reaction between mPC and SAG. Blends of PC (or mPC) and SAN with a ratio of 20:80 were melt mixed with 5 wt% SGMA. Fig. 7 illustrates the phase morphologies obtained from this process. Notably, SEM images clearly exhibit a substantial reduction in the domain size of PC within the blend containing modified PC compared to that of unmodified PC ($R_v = 0.98 \mu\text{m}$). The obtained results clearly demonstrate the significant role played by the OH groups present in mPC in facilitating the reaction with SAG and thereby promoting the formation of PC-SAG copolymers, as discussed earlier. These newly formed copolymers act as effective compatibilizers for the PC and SAN blend. Even when water-modified PC is used, there is a noticeable decrease in domain size ($R_v = 0.30 \mu\text{m}$), indicating a compatibilization effect. However, the degree of compatibilization is lower compared to when NaOH is added along with the same amount of water ($R_v = 0.19 \mu\text{m}$). This can be attributed to the lower concentration of OH groups in the water-modified PC. It is worth noting that during specimen fracture, plastic flow was observed around the domains, indicating robust interfacial adhesion. This robust adhesion is promoted by the copolymers formed through the reactive coupling of PC and SAG. Furthermore, Fig. 8 illustrates our confirmation of the presence of Na ion aggregates through TEM and EDX analysis of mPC/SAN/SAG samples.

To further investigate the potential of modified PC, blends of PC and mPC with varying ratios (90/10 and 80/20) were prepared, and their tensile properties were evaluated. Fig. 9 illustrates the results, indicating a decrease in modulus compared to pure PC. However, it is noteworthy that the tensile elongation significantly increased, especially with a blend ratio of 90/10, where it surpassed three times the elongation of pure PC. Additionally, the melt viscosities of the blends were found to be lower than those of pure PC. However, it is noted that the melt viscosities of the 80/20 blend are slightly higher than those of the 90/10 blend, as depicted in Fig. 10. The viscosity of the PC/mPC blend does not seem to decrease with increasing mPC content, indicating a complex phenomenon likely influenced by ionic aggregates. Nevertheless, these unique characteristics observed in Figs. 9 and 10 offer distinct advantages for applications that require high flowability during molding processes. By adjusting the proportion of mPC appropriately, it becomes possible to control both the mechanical properties and processability of PC without compromising its transparency.

The significant increase in modulus, tear, and abrasion resistance observed in ionomers compared to their nonionic parent polymers is a well-established fact [22–24]. These improved properties in ionomers are attributed to the aggregation of ionic groups into microdomains. In accordance with this understanding, one plausible explanation for the enhanced elongation of the PC/mPC blend could be linked to the influence of ion aggregates within the PC matrix. However, the data presented in Fig. 8 indicate a decrease in modulus for the blend containing Na ions, which contradicts the behavior typically observed in ionomers. As a result, further research is necessary to fully comprehend the role of Na ion aggregations in the deformation behavior of such materials. Additional investigation will provide valuable insights into these systems and help elucidate the underlying mechanisms at play.

4. Conclusions

The controlled degradation of PC was carried out by incorporating different concentrations of NaOH aqueous solution during the melt processing in a batch mixer. The results revealed that the addition of NaOH caused a much larger change in the melt viscosity of PC compared to when only pure water was added. In the case of water addition alone, the melt viscosity of the polymer decreased. However, when NaOH was added above a certain concentration, the modification resulted in an increase in the melt viscosity of the polymer.

The increase in NaOH concentration was observed to directly correlate with an increase in the concentration of OH groups generated at the ends of PC molecules. These functional groups effectively reacted

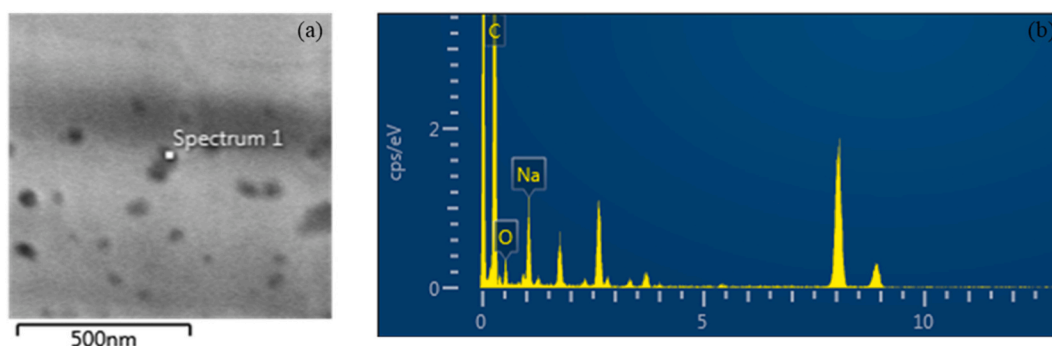


Fig. 8. TEM image (a) and EDX spectrum (b) for mPC/SAN/SAG (same composition as Fig. 7(c)).

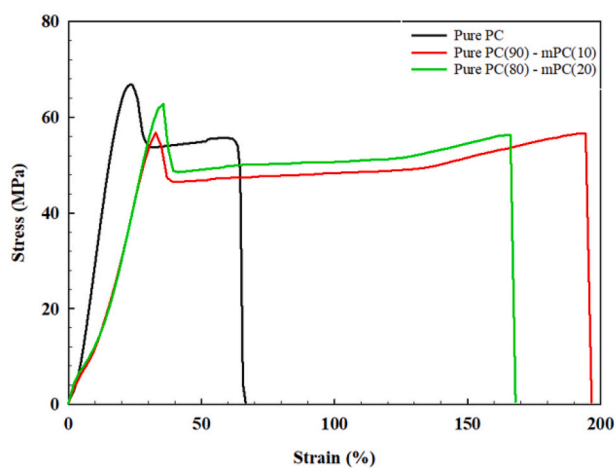


Fig. 9. Tensile properties of PC and blends of PC/mPC.

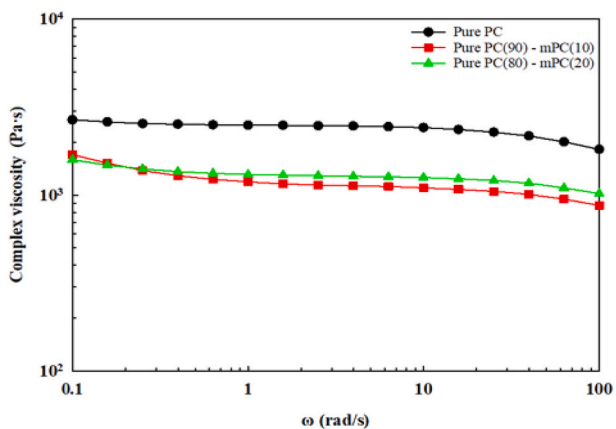


Fig. 10. Changes of complex viscosity of PC when blended with mPC (10 wt% and 20 wt%, respectively) at 240 °C.

with SAG, thereby improving the compatibility of the PC/SAN/SAG blend. Additionally, when 10 wt% of the modified PC was blended with PC, the melt viscosity of PC decreased while the elongation increased more than three times. The modification of PC using an appropriate concentration of NaOH aqueous solution proves to be a useful method for controlling the rheological and mechanical properties, as well as functionalizing the end groups of PC resin. However, further research is required to more clearly elucidate the structure of Na ion aggregates and their impact on the deformation behavior of the modified PC.

Author statement

We confirm that the manuscript has been read and approved by all named authors and that there are no other persons who satisfied the criteria for authorship but are not listed. We further confirm that the order of authors listed in the manuscript has been approved by all of us.

CRediT author statements

Choi Y: Experiment, Investigation, Writing-original draft preparation. Kim H: Conceptualization, Supervision, Resources, Writing review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgments

This research was supported by Basic Science Research Program through the National Research Foundation of Korea (NRF) funded by the Ministry of Education, Science and Technology (NRF-2019R1I1A2A01051610).

References

- [1] C. Tzoganakis, Y. Yang, J. Vlachopoulos, A.E. Hamielec, Controlled degradation of polypropylene: a comprehensive experimental and theoretical investigation, *Polym.-Plast Technol* 28 (1989) 319–350.
- [2] H.G. Fritz, B. Stöhrer, Polymer compounding process for controlled peroxide degradation of PP, *Int. Polym. Process.* 1 (1986) 31–41.
- [3] Y. Ito, H. Ogasawara, Y. Ishida, H. Ohtani, S. Tsuge, Characterization of end groups in polycarbonates by reactive pyrolysis-gas chromatography, *Polym. J.* 28 (1996) 1090–1095.
- [4] C.A. Pryde, M.Y. Hellman, Solid state hydrolysis of bisphenol-A polycarbonate. I. Effect of phenolic groups, *J. Appl. Polym. Sci.* 25 (1980) 2573–2587.
- [5] H.E. Bair, D.R. Falcone, M.Y. Hellman, G.E. Johnson, P.G. Kelleher, Hydrolysis of polycarbonate to yield BPA, *J. Appl. Polym. Sci.* 26 (1981) 1777–1786.
- [6] F.C. Schilling, W.M. Ringo, N.J.A. Sloane, F.A. Bovey, Carbon-13 nuclear magnetic resonance study of the hydrolysis of bisphenol A polycarbonate, *Macromolecules* 14 (1981) 532–537.
- [7] I. Ghorbel, N. Akele, F. Thominet, P. Spiteri, J. Verdu, Hydrolytic aging of polycarbonate. II. Hydrolysis kinetics, Effect of static stresses, *J. Appl. Polym. Sci.* 55 (1995) 173–179.
- [8] E.V. Antonakou, D.S. Achilias, Recent advances in polycarbonate recycling: a review of degradation methods and their mechanisms, *Waste Biomass Valor* 4 (2013) 9–21.
- [9] R. Piñero, J. Garcia, M.J. Cocero, Chemical recycling of polycarbonate in a semi-continuous lab-plant. A green route with methanol and methanol-water mixtures, *Green Chem.* 7 (2005) 380–387.

- [10] F. Liu, Z. Li, S. Yu, X. Cui, C. Xie, X. Ge, Methanolysis and hydrolysis of polycarbonate under moderate conditions, *J. Polym. Environ.* 17 (2009) 208–211.
- [11] X. Song, F. Liu, L. Li, X. Yang, S. Yu, X. Ge, Hydrolysis of polycarbonate catalyzed by ionic liquid [Bmim][Ac], *J. Hazard Mater.* 244–245 (2013) 204–208.
- [12] W. Hale, H. Keskkula, D.R. Paul, Compatibilization of PBT/ABS blends by methyl methacrylate-glycidyl methacrylate-ethyl acrylate terpolymers, *Polymer* 40 (1999) 365–377.
- [13] P.C. Lee, W.F. Kuo, F.C. Chang, In situ compatibilization of PBT/ABS blends through reactive copolymers, *Polymer* 35 (1994) 5641–5650.
- [14] J.W. Lee, S.C. Lee, H. Kim, Effect of mixing protocol on the morphology of multi-walled carbon nanotube incorporated ternary blends of polycarbonate, styrene-acrylonitrile, and poly(methyl methacrylate) in a twin screw extruder, *Korea Aust. Rheol. J.* 33 (2021) 143–150.
- [15] R.S. McLean, M. Doyle, B.B. Sauer, High-resolution imaging of ionic domains and crystal morphology in ionomers using AFM techniques, *Macromolecules* 33 (2000) 6541–6550.
- [16] S. Trinkle, C. Friedrich, Van Gorp-Palmen-plot: a way to characterize polydispersity of linear polymers, *Rheol. Acta* 40 (2001) 322–328.
- [17] S. Trinkle, P. Walter, C. Friedrich, Van Gorp-Palmen plot II - classification of long chain branched polymers by their topology, *Rheol. Acta* 41 (2002) 103–113.
- [18] D. Auhl, J. Stange, H. Münstedt, B. Krause, D. Voigt, A. Lederer, U. Lappan, K. Lunkwitz, Long-chain branched polypropylenes by electron beam irradiation and their rheological properties, *Macromolecules* 37 (2004) 9465–9472.
- [19] R. Lagendijk, A.H. Hogt, A. Buijtenhuijs, A.D. Gotsis, Peroxydicarbonate modification of polypropylene and extentional flow properties, *Polymer* 42 (2001) 10035–10043.
- [20] R.S. McLean, M. Doyle, B.B. Sauer, High-resolution imaging of ionic domains and crystal morphology in ionomers using AFM techniques, *Macromolecules* 33 (2000) 6541–6550.
- [21] G.L. Gaines, Acceleration of hydrolysis of bisphenol-A polycarbonate by hindered amines, *Polym. Degrad. Stabil.* 27 (1990) 13–18.
- [22] W.J. MacKnight, T.R. Earnest Jr., The structure and properties of ionomers, *Macromol. Rev.* 16 (1981) 41–122.
- [23] A. Eisenberg, Clustering of ions in organic polymers. A theoretical approach, *Macromolecules* 3 (1970) 147–154.
- [24] M. Hara, J.A. Sauer, Mechanical properties of ionomers, *Rev Macromol Chem Phys* C34 (1994) 325–373.